

Binary to Decimal Conversion

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1

Binary is either set to 1 or 0 | on or off

Example Sting: 01101101

128	64	32	16	8	4	2	1
0	1	1	0	1	1	0	1
0	64	32	0	8	4	0	1

$$64+32+8+4+1 = 109$$

11110000 00110111 11001100 00000011 (this example can be an ip string in multiple octets.

1111|0000 0011|0111 1100|1100 0000|0011 (Separate into groups of four to make math easier

$$1111|0000 = 128+64+32+16 = 240$$

$$0011|0111 = 32+16+4+2+1 \text{ or } 48 + 7 = 55$$

$$1100|1100 = 128+64+8+4 \text{ } 192+8+4 = 204$$

$$0000|0011 = 2+1 = 3$$

(11110000 00110111 11001100 00000011) is 240.55.204.3

Decimal to Binary Conversion

Essentially just the opposite of above, start with a decimal and then work it out into binary using the chart below

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1

Use the blank cells in row 3 to mark which numbers will be in the additive equation to solve for X
X being the decimal value you want to write in Binary

Decimal Value 217

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1
1	1	0	1	1	0	0	1

217 is less than 240, so the first half of the octet is not full (ie all 1's or 1111)

$$128+64+32 = 192 + 32 = 224$$

224 is higher than 217, so 32 will not be included

$$128+64+16 = 192+16 = 208 \text{ (here have the first half of our octet - 1101)}$$

$$217-208 \text{ (first half of octet)} = 9 \text{ (as such 9 is the second half of our octet - 1001)}$$

$$217 = 208 + 9 \text{ or binary } 11011001$$

$$192.168.1.64 / 26$$

Break it up with 8 hosts per subnet

8 hosts

$$8 = 2^n - 2$$

$$8 = 2^4 - 2$$

$$16 - 2 = 14$$

$$192.168.1.64 / 26$$

8.8.8. 11 00 0000

N = 4, so steal 4 bit

$$192.168.1.64 / 26$$

128 is the next sub net

127 is the broadcast address

126 is the final host

64 = 0100 0000 -is the subnet address and 65 is the first host

/26

/28

8.8.8. 01 000000 192.168.1.64 /29

8.8.8. 01 001000 192.168.1.80 /29

8.8.8. 01 011 0000 192.168.1.96 /29

8.8.8. 01 111 0000 192.168.1.112 /29

192.168.1.64 0000/ 29

8.8.8. 1111.1 00064

64 = 01 00 0000

/

192.168.1.01 11 100